Criblage DFT (WB97X-D4/aug-cc-pVTZ) des allotropes polyazotés N_x : rapport d'étape

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Abstract

Ce document rend compte de la campagne DFT en cours au niveau WB97X-D4/aug-cc-pVTZ sur les **193 candidats** présélectionnés par criblage GFN2-xTB (N_4 à N_{16}, familles neutre/cation/anion). État au moment de la rédaction : **92 structures optimisées et vérifiées par calcul de fréquences** (48 %), 0 erreur ORCA, campagne active sur cluster SLURM. Une confrontation aux bases de données et à la littérature déjà compilées dans `polyN-pipeline` montre que, parmi les 7 groupes (formule, charge) où une comparaison est possible, le criblage local a déjà trouvé un isomère plus stable que tout ce qui est catalogué dans 4 d'entre eux (N_9^-, N_9^+, N_{10}, N_{12}), tandis que 165 des 193 candidats (86 %) ne correspondent à aucune topologie référencée ailleurs. Les deux ancres de référence pour les ΔH_f ioniques (N_5^- cyclique, N_5^+ coudé) ont été retrouvées dans `biblio_polyN/final_report_isodesmic.csv` et permettent désormais de calculer un ΔH_f absolu (niveau GFN2-xTB) pour les 193 candidats – tables 2–4.

Contents

1. Méthode

1.1 Origine des candidats

Les 193 structures de départ proviennent du criblage GFN2-xTB de `polyN-pipeline` (génération RDKit ETKDGv3 + énumération de graphes `geng`/nauty + substitution isolobale C→N + échantillonnage aléatoire, selon la formule), sélection par stoechiométrie et vérification de fréquences à ce niveau. Aucune contribution du second dépôt compagnon `polyN_adapt` (génération adaptative/surrogate) n'a été identifiée dans ce jeu – voir §2, à confirmer. Les formules à N impair (N_5, N_7, N_9, N_{11}, N_{13}, N_{15}) ne sont représentées qu'en cation/anion (contrainte de parité électronique), les formules paires principalement en neutre.

1.2 Niveau de calcul

Optimisation de géométrie puis fréquences, WB97X-D4/aug-cc-pVTZ, TightSCF, DEFGRID3 (ORCA 6.1.0). Deux protocoles selon la taille :

- **N_4–N_{11}** (« petits ») : un job unique Opt+Freq, sans contrainte de symétrie – évite tout risque de figer un candidat Jahn-Teller dans un point-selle de symétrie artificiellement élevée.
- **N_{12}–N_{16}** (« gros ») : chaîne en trois étapes Opt (sans symétrie) → ré-optimisation rapide avec `UseSym` (nettoie la géométrie vers le groupe ponctuel exact) → Freq avec `UseSym` (diagonalisation du Hessien exploitant la symétrie complète, pas seulement les sous-groupes de D_{2h}). Surcoût mesuré d'un facteur ~ 2 en temps de calcul sur un cas test (N_5^-, 7:41 → 15:35), jugé acceptable uniquement pour les grosses structures.

Distribution sur cluster SLURM via un ordonnanceur maison (`production_dispatch.py`) : surveillance en continu des cœurs libres sur les nœuds alloués, soumission automatique du prochain candidat dès qu'un créneau se libère, 8 cœurs par job (optimum mesuré : efficacité CPU 75–96 %, gain marginal au-delà ; 24–32 cœurs pour les gros tant que la file correspondante n'était pas épuisée, efficacité 90–96 % mesurée jusqu'à 32 cœurs sur N_{12}).

Catégorie	n	Wall moy. (min)	Wall min (min)	Wall max (min)	CPU-h moy.
Petits (8 cœurs, N4–N11)	83	72.4	5.1	262.9	8.88
Gros (24/32 cœurs, N12–N16)	23	116.9	23.3	310.4	46.52
Total	106	82.1	5.1	310.4	17.05

Table 1: Temps de calcul (écoulé = wall-clock, CPU = temps utilisateur+système cumulé sur tous les cœurs) sur les 106 jobs terminés au moment de la rédaction.

2. Provenance et statut de chaque candidat

C'est la table centrale de ce rapport : pour chacun des 193 candidats, son origine, son statut DFT actuel, son ΔH_f GFN2-xTB (ancré sur les valeurs littérature CCSD(T) retrouvées dans `biblio_polyN/final_report_isodesmic.csv` – $-106{,}4$ kcal/mol pour le pentagone N_5^- (D_{5h}), $195{,}9$ kcal/mol pour le N_5^+ coudé/Vchain (C_{2v}, valeur expérimentale par diffraction X concordante à $195{,}8$), $\Delta H_f(N_2) = 0$ pour les neutres par construction) et, quand une comparaison est possible, la topologie cataloguée à laquelle il correspond.

Les deux ancres ont été recalculées indépendamment ici (GFN2-xTB, xtb 6.7.1) et reproduisent les valeurs internes de `polyN_pipeline.py` à 5–6 décimales : $E(N_2) = -5{,}763935$ Ha,

$E(\mathrm{N}_5^-$, D_{5h}, = `N5_anion_anion_001`$) = -14{,}760899$ Ha, $E(\mathrm{N}_5^+$, C_{2v}, = `N5_cation_cation_001`$) = -13{,}985267$ Ha. Point notable : la référence interne du pipeline pour la famille cation n'est pas ce dernier mais `N5_cation_cation_004` (isomère C_1 différent, $E = -13{,}816935$ Ha, $105{,}6$ kcal/mol moins stable) – les ΔH ci-dessous sont recalés sur le C_{2v} qui correspond à la structure littérature.

Trois tableaux distincts (neutres, cations, anions), triés par nombre d'atomes croissant puis par stabilité au sein de chaque formule ($\Delta H = 0$ pour l'isomère le plus stable, DFT quand disponible sinon GFN2-xTB). Chaque ligne étant un de nos 193 candidats, *our work* s'applique toujours ; quand la topologie correspond en plus à un article publié, son numéro est cité à côté ("`[k], our work`" – liste des articles en Annexe B). Sans correspondance article, seul *our work* apparaît – ce qui inclut, par choix explicite, les correspondances aux bases externes `N_csp` et `polyN_study` (candidats, pas des articles publiés) aussi bien que les topologies réellement inédites.

Attention sur la portée de cette bibliographie : les 22 références numérotées ici ([GS], [B], [A1]–[A20]) sont celles pour lesquelles `polyN-pipeline` a déjà extrait une structure et recalculé son ΔH_f isodesmique (110 structures nommées dans `table_isodesmic_{neutre,anion,cation}.tex`). Le dépôt contient séparément une recherche de citations bien plus large (135 articles en génération 1, `table_citation_search.tex` + `_gen2.tex`) qui n'est **pas encore** réduite en structures nommées avec référence – ces papiers supplémentaires ne peuvent donc pas être cités par candidat tant que ce travail d'extraction n'est pas fait. Une première passe de correction ici (priorité à une correspondance biblio_article existante plutôt qu'à un doublon de base externe de même topologie mais moindre énergie) a fait passer les correspondances de 1 à 8 candidats.

Table 2: Composés **neutres**, classés par nombre d'atomes croissant puis par stabilité (isomère le plus stable = $\Delta H = 0$). ΔH au niveau DFT quand disponible, sinon GFN2-xTB (indiqué en colonne). "$> 1{,}7$" : la structure contient au moins une distance N–N supérieure à 1,7 Å. Référence : numéro d'article (voir Annexe Références) ou *our work* si absente de la biblio actuelle.

Nom	N	PG	ΔH (kcal/mol)	niveau	$> 1{,}7$Å	Réf.
`N4_neutral_neutral_001`	4	Td	0.00	DFT		our work
`N4_neutral_neutral_004`	4	D2h	3.38	DFT		our work
`N4_neutral_neutral_002`	4	C2v	16.85	DFT		our work
`N4_neutral_neutral_003`	4	C2v	33.07	DFT		our work
`N6_neutral_neutral_004`	6	C1	0.00	DFT		our work
`N6_neutral_neutral_005`	6	–	64.62	xtb		our work
`N6_neutral_neutral_011`	6	–	119.57	xtb		our work
`N6_neutral_neutral_001`	6	C2	172.76	DFT		our work

(suite)

N6_neutral_neutral_002	6	D2	195.83	DFT	[1], our work
N6_neutral_neutral_003	6	C1	200.83	DFT	our work
N6_neutral_neutral_007	6	C2	226.12	DFT	our work
N6_neutral_neutral_009	6	C2	231.01	DFT	our work
N6_neutral_neutral_008	6	C2v	236.66	DFT	our work
N6_neutral_neutral_006	6	C1	258.25	DFT	our work
N6_neutral_neutral_012	6	D3h	305.23	DFT	our work
N6_neutral_neutral_010	6	C2v	322.55	DFT	our work
N8_neutral_neutral_001	8	–	0.00	xtb	[1], our work
N8_neutral_neutral_002	8	–	9.91	xtb	[2], our work
N8_neutral_neutral_003	8	–	20.26	xtb	[1], our work
N8_neutral_neutral_004	8	–	28.09	xtb	[1], our work
N8_neutral_neutral_005	8	–	29.41	xtb	[1], our work
N8_neutral_neutral_006	8	–	56.53	xtb	our work
N8_neutral_neutral_007	8	–	66.38	xtb	our work
N8_neutral_neutral_008	8	–	67.22	xtb	our work
N8_neutral_neutral_009	8	–	73.70	xtb	our work
N8_neutral_neutral_010	8	–	84.06	xtb	our work
N8_neutral_neutral_011	8	–	84.76	xtb	our work
N8_neutral_neutral_012	8	–	93.97	xtb	our work
N8_neutral_neutral_013	8	–	102.90	xtb	our work
N8_neutral_neutral_014	8	–	113.23	xtb	our work
N8_neutral_neutral_015	8	–	126.26	xtb	our work
N8_neutral_neutral_016	8	–	132.78	xtb	our work
N8_neutral_neutral_017	8	–	139.04	xtb	our work

(suite)

N8_neutral_neutral_018	8	–	140.21	xtb	our work
N8_neutral_neutral_019	8	–	150.30	xtb	our work
N8_neutral_neutral_020	8	–	152.93	xtb	our work
N8_neutral_neutral_021	8	–	155.66	xtb	our work
N8_neutral_neutral_022	8	–	161.85	xtb	our work
N8_neutral_neutral_023	8	–	164.68	xtb	our work
N8_neutral_neutral_024	8	–	183.05	xtb	our work
N8_neutral_neutral_025	8	–	205.70	xtb	our work
N10_neutral_neutral_001	10	C2	0.00	DFT	our work
N10_neutral_neutral_002	10	C1	30.61	DFT	our work
N10_neutral_neutral_003	10	C1	50.60	DFT	[1], our work
N10_neutral_neutral_008	10	–	59.29	xtb	our work
N10_neutral_neutral_009	10	–	63.47	xtb	our work
N10_neutral_neutral_006	10	Cs	69.56	DFT	our work
N10_neutral_neutral_004	10	C1	71.13	DFT	[1], our work
N10_neutral_neutral_007	10	Cs	87.34	DFT	our work
N10_neutral_neutral_012	10	–	98.78	xtb	our work
N10_neutral_neutral_005	10	C1	102.77	DFT	our work
N10_neutral_neutral_011	10	C1	107.23	DFT	our work
N10_neutral_neutral_010	10	C1	118.99	DFT	our work
N10_neutral_neutral_015	10	C2	123.40	DFT	our work
N10_neutral_neutral_016	10	C1	126.63	DFT	our work
N10_neutral_neutral_014	10	C1	139.29	DFT	our work
N10_neutral_neutral_017	10	C2v	151.14	DFT	our work
N10_neutral_neutral_021	10	C2v	173.84	DFT	our work

(suite)

N10_neutral_neutral_013	10	C2v	193.75	DFT	our work
N10_neutral_neutral_018	10	C2	199.07	DFT	our work
N10_neutral_neutral_019	10	C3v	203.80	DFT	our work
N10_neutral_neutral_020	10	C2	214.21	DFT	our work
N10_neutral_neutral_022	10	D5h	241.14	DFT	our work
N12_neutral_neutral_001	12	C1	0.00	DFT	our work
N12_neutral_neutral_003	12	C1	36.41	DFT	our work
N12_neutral_neutral_002	12	C1	56.99	DFT	our work
N12_neutral_neutral_004	12	Cs	116.12	DFT	our work
N12_neutral_neutral_006	12	C1	127.50	DFT	our work
N12_neutral_neutral_005	12	C1	138.12	DFT	our work
N12_neutral_neutral_007	12	D3d	188.33	DFT	our work
N12_neutral_neutral_008	12	C1	219.30	DFT	our work
N12_neutral_neutral_009	12	D6h	334.92	DFT	our work
N14_neutral_neutral_004	14	C1	0.00	DFT	our work
N14_neutral_neutral_002	14	–	40.76	xtb	our work
N14_neutral_neutral_003	14	–	77.89	xtb	our work
N14_neutral_neutral_001	14	C1	250.59	DFT	our work
N14_neutral_neutral_005	14	C2	341.35	DFT	our work
N16_neutral_neutral_001	16	–	0.00	xtb	our work
N16_neutral_neutral_002	16	–	36.70	xtb	our work

Table 3: Composés **cationiques**, classés par nombre d'atomes croissant puis par stabilité (isomère le plus stable = $\Delta H = 0$). ΔH au niveau DFT quand disponible, sinon GFN2-xTB (indiqué en colonne). "$> 1{,}7$" : la structure contient au moins une distance N–N supérieure à 1,7 Å. Référence : numéro d'article (voir Annexe Références) ou *our work* si absente de la biblio actuelle.

Nom	**N**	**PG**	ΔH **(kcal/mol)**	**niveau**	$> 1{,}7$**Å**	**Réf.**
N5_cation_cation_001	5	C2v	0.00	DFT		our work
N5_cation_cation_004	5	C1	0.01	DFT		our work
N5_cation_cation_002	5	Cs	41.63	DFT		our work
N5_cation_cation_003	5	–	104.85	xtb		our work
N5_cation_cation_005	5	D2d	109.53	DFT		our work
N5_cation_cation_006	5	C4v	155.74	DFT		our work
N5_cation_cation_007	5	–	185.58	xtb		our work
N7_cation_cation_001	7	Cs	0.00	DFT		our work
N7_cation_cation_002	7	–	6.20	xtb		our work
N7_cation_cation_003	7	–	27.01	xtb		our work
N7_cation_cation_004	7	–	91.26	xtb		our work
N7_cation_cation_005	7	–	124.94	xtb		our work
N7_cation_cation_006	7	–	178.83	xtb		our work
N9_cation_cation_001	9	C1	0.00	DFT		our work
N9_cation_cation_002	9	–	29.44	xtb		our work
N9_cation_cation_003	9	–	31.97	xtb		our work
N9_cation_cation_004	9	–	34.20	xtb		our work
N9_cation_cation_005	9	–	100.60	xtb		our work
N9_cation_cation_006	9	–	104.54	xtb		our work
N9_cation_cation_007	9	–	113.87	xtb		our work

(suite)

N9_cation_cation_008	9	–	114.89	xtb	our work
N9_cation_cation_009	9	–	117.35	xtb	our work
N9_cation_cation_010	9	–	132.31	xtb	our work
N9_cation_cation_011	9	–	174.16	xtb	our work
N9_cation_cation_012	9	–	176.85	xtb	our work
N11_cation_cation_002	11	C1	0.00	DFT	our work
N11_cation_cation_001	11	C2v	23.30	DFT	our work
N11_cation_cation_005	11	C2	45.85	DFT	our work
N11_cation_cation_003	11	–	46.08	xtb	our work
N11_cation_cation_004	11	–	61.37	xtb	our work
N11_cation_cation_006	11	C1	67.68	DFT	our work
N11_cation_cation_007	11	–	128.07	xtb	our work
N11_cation_cation_008	11	–	195.40	xtb	our work
N11_cation_cation_009	11	–	239.73	xtb	our work
N11_cation_cation_010	11	–	239.73	xtb	our work
N12_cation_cation_001	12	C1	0.00	DFT	our work
N13_cation_cation_001	13	C1	0.00	DFT	our work
N13_cation_cation_002	13	–	40.46	xtb	our work
N13_cation_cation_003	13	–	56.53	xtb	our work
N13_cation_cation_004	13	C1	151.22	DFT	our work
N15_cation_cation_001	15	C1	0.00	DFT	our work
N15_cation_cation_002	15	C1	327.87	DFT	our work

Table 4: Composés **anioniques**, classés par nombre d'atomes croissant puis par stabilité (isomère le plus stable = $\Delta H = 0$). ΔH au niveau DFT quand disponible, sinon GFN2-xTB (indiqué en colonne). "$> 1{,}7$" : la structure contient au moins une distance N–N supérieure à 1,7 Å. Référence : numéro d'article (voir Annexe Références) ou *our work* si absente de la biblio actuelle.

Nom	N	PG	ΔH (kcal/mol)	niveau	$> 1{,}7$Å	Réf.
N5_anion_anion_002	5	C1	0.00	DFT		our work
N5_anion_anion_003	5	C1	0.00	DFT		our work
N5_anion_anion_004	5	C1	0.00	DFT		our work
N5_anion_anion_001	5	D5h	6.58	DFT		our work
N5_anion_anion_007	5	C1	70.42	DFT		our work
N5_anion_anion_005	5	C1	70.42	DFT		our work
N5_anion_anion_008	5	C1	72.16	DFT		our work
N5_anion_anion_006	5	Cs	111.04	DFT		our work
N5_anion_anion_009	5	–	119.67	xtb		our work
N5_anion_anion_010	5	C2v	177.35	DFT		our work
N7_anion_anion_001	7	–	0.00	xtb		our work
N7_anion_anion_011	7	C1	0.00	DFT		our work
N7_anion_anion_002	7	–	2.77	xtb		our work
N7_anion_anion_012	7	C1	3.08	DFT		our work
N7_anion_anion_003	7	–	7.94	xtb		our work
N7_anion_anion_004	7	–	9.12	xtb		our work
N7_anion_anion_016	7	C1	14.43	DFT		our work
N7_anion_anion_010	7	C2v	19.98	DFT		our work
N7_anion_anion_005	7	–	22.23	xtb		our work
N7_anion_anion_008	7	C1	23.07	DFT		our work

(suite)

N7_anion_anion_006	7	–	30.87	xtb	our work
N7_anion_anion_007	7	–	41.65	xtb	our work
N7_anion_anion_009	7	–	47.46	xtb	our work
N7_anion_anion_018	7	Cs	53.06	DFT	our work
N7_anion_anion_013	7	–	54.11	xtb	our work
N7_anion_anion_014	7	–	59.93	xtb	our work
N7_anion_anion_019	7	C1	63.44	DFT	our work
N7_anion_anion_015	7	–	71.61	xtb	our work
N7_anion_anion_017	7	–	82.01	xtb	our work
N7_anion_anion_020	7	–	144.93	xtb	our work
N9_anion_anion_001	9	C1	0.00	DFT	our work
N9_anion_anion_002	9	–	19.37	xtb	our work
N9_anion_anion_003	9	–	23.62	xtb	our work
N9_anion_anion_004	9	–	29.47	xtb	our work
N9_anion_anion_005	9	–	33.08	xtb	our work
N9_anion_anion_006	9	–	35.95	xtb	our work
N9_anion_anion_007	9	–	46.38	xtb	our work
N9_anion_anion_008	9	–	53.91	xtb	our work
N9_anion_anion_009	9	–	55.92	xtb	our work
N9_anion_anion_010	9	–	57.07	xtb	our work
N9_anion_anion_011	9	–	60.00	xtb	our work
N9_anion_anion_012	9	–	86.79	xtb	our work
N9_anion_anion_013	9	–	87.40	xtb	our work
N9_anion_anion_014	9	–	88.41	xtb	our work
N9_anion_anion_015	9	–	104.99	xtb	our work

(suite)

N9_anion_anion_016	9	–	126.07	xtb	our work
N9_anion_anion_017	9	–	128.05	xtb	our work
N9_anion_anion_018	9	–	141.31	xtb	our work
N9_anion_anion_019	9	–	153.96	xtb	our work
N9_anion_anion_020	9	–	205.40	xtb	our work
N11_anion_anion_001	11	C1	0.00	DFT	our work
N11_anion_anion_002	11	C1	38.48	DFT	our work
N11_anion_anion_006	11	C1	60.32	DFT	our work
N11_anion_anion_011	11	C1	74.02	DFT	our work
N11_anion_anion_005	11	C1	74.39	DFT	our work
N11_anion_anion_009	11	–	79.30	xtb	our work
N11_anion_anion_004	11	C1	92.16	DFT	our work
N11_anion_anion_007	11	Cs	101.73	DFT	our work
N11_anion_anion_010	11	Cs	102.27	DFT	our work
N11_anion_anion_008	11	C1	108.61	DFT	our work
N11_anion_anion_012	11	–	108.72	xtb	our work
N11_anion_anion_003	11	C2	109.19	DFT	our work
N11_anion_anion_014	11	–	119.70	xtb	our work
N11_anion_anion_015	11	–	146.04	xtb	our work
N11_anion_anion_016	11	–	164.02	xtb	our work
N11_anion_anion_013	11	C1	177.01	DFT	our work
N11_anion_anion_017	11	C1	185.47	DFT	our work
N11_anion_anion_018	11	C1	199.63	DFT	our work
N13_anion_anion_001	13	C1	0.00	DFT	our work

(suite)

N13_anion_anion_002	13	–	115.00	xtb	our work
N13_anion_anion_003	13	–	116.61	xtb	our work
N13_anion_anion_004	13	–	123.73	xtb	our work

3. Classement de stabilité : DFT contre xTB

Les tableaux suivants classent chaque isomère par rang DFT croissant (rang 1 = ground state DFT) au sein de sa formule, avec le rang xTB d'origine rappelé en regard – **la question posée est de savoir si le criblage xTB, nettement moins cher, prédit le même classement de stabilité que le DFT**, ou si des réordonnancements significatifs apparaissent.

Table 5: Classement DFT vs xTB, composés **neutres**. Trié par rang DFT (ground state DFT = rang 1) au sein de chaque formule ; le rang xTB tel que généré à l'origine est rappelé pour voir si les deux classements concordent.

Nom	N	rang xtb	rang DFT	E_{rel} xtb (kcal/mol)	E_{rel} DFT (kcal/mol)	accord GS
N4_neutral_neutral_001	4	1	1	0.0	0.0	
N4_neutral_neutral_004	4	4	2	41.64	3.375	
N4_neutral_neutral_002	4	2	3	26.47	16.846	
N4_neutral_neutral_003	4	3	4	31.01	33.074	
N6_neutral_neutral_004	6	4	1	60.59	0.0	
N6_neutral_neutral_001	6	1	2	0.0	172.762	
N6_neutral_neutral_002	6	2	3	24.09	195.834	
N6_neutral_neutral_003	6	3	4	55.0	200.828	
N6_neutral_neutral_007	6	7	5	83.63	226.124	
N6_neutral_neutral_009	6	9	6	110.8	231.008	
N6_neutral_neutral_008	6	8	7	84.52	236.662	
N6_neutral_neutral_006	6	6	8	78.8	258.252	
N6_neutral_neutral_012	6	12	9	136.66	305.226	
N6_neutral_neutral_010	6	10	10	114.33	322.554	
N10_neutral_neutral_001	10	1	1	0.0	0.0	
N10_neutral_neutral_002	10	2	2	10.5	30.609	
N10_neutral_neutral_003	10	3	3	23.5	50.598	
N10_neutral_neutral_006	10	6	4	38.56	69.561	
N10_neutral_neutral_004	10	4	5	33.36	71.134	
N10_neutral_neutral_007	10	7	6	53.65	87.344	
N10_neutral_neutral_005	10	5	7	36.56	102.766	
N10_neutral_neutral_011	10	11	8	92.95	107.23	
N10_neutral_neutral_010	10	10	9	88.62	118.993	
N10_neutral_neutral_015	10	15	10	111.24	123.395	
N10_neutral_neutral_016	10	16	11	113.83	126.633	
N10_neutral_neutral_014	10	14	12	100.5	139.293	
N10_neutral_neutral_017	10	17	13	130.46	151.139	

(suite)

Nom	N	rang xtb	rang DFT	E_{rel} xtb	E_{rel} DFT	accord GS
N10_neutral_neutral_021	10	21	14	174.6	173.835	
N10_neutral_neutral_013	10	13	15	99.55	193.748	
N10_neutral_neutral_018	10	18	16	143.02	199.068	
N10_neutral_neutral_019	10	19	17	147.97	203.805	
N10_neutral_neutral_020	10	20	18	168.46	214.208	
N10_neutral_neutral_022	10	22	19	183.94	241.136	
N12_neutral_neutral_001	12	1	1	0.0	0.0	
N12_neutral_neutral_003	12	3	2	35.91	36.408	
N12_neutral_neutral_002	12	2	3	13.24	56.987	
N12_neutral_neutral_004	12	4	4	45.58	116.12	
N12_neutral_neutral_006	12	6	5	76.36	127.5	
N12_neutral_neutral_005	12	5	6	68.53	138.121	
N12_neutral_neutral_007	12	7	7	100.96	188.328	
N12_neutral_neutral_008	12	8	8	133.16	219.304	
N12_neutral_neutral_009	12	9	9	242.43	334.919	
N14_neutral_neutral_004	14	4	1	101.56	0.0	
N14_neutral_neutral_001	14	1	2	0.0	250.586	
N14_neutral_neutral_005	14	5	3	133.34	341.354	

Table 6: Classement DFT vs xTB, composés **cationiques**. Trié par rang DFT (ground state DFT = rang 1) au sein de chaque formule ; le rang xTB tel que généré à l'origine est rappelé pour voir si les deux classements concordent.

Nom	N	rang xtb	rang DFT	E_{rel} xtb (kcal/mol)	E_{rel} DFT (kcal/mol)	accord GS
N5_cation_cation_001	5	1	1	0.0	0.0	
N5_cation_cation_004	5	4	2	105.63	0.005	
N5_cation_cation_002	5	2	3	74.16	41.633	
N5_cation_cation_005	5	5	4	134.0	109.534	
N5_cation_cation_006	5	6	5	184.33	155.736	
N7_cation_cation_001	7	1	1	0.0	0.0	
N9_cation_cation_001	9	1	1	0.0	0.0	
N11_cation_cation_002	11	2	1	2.67	0.0	
N11_cation_cation_001	11	1	2	0.0	23.301	
N11_cation_cation_005	11	5	3	73.62	45.845	
N11_cation_cation_006	11	6	4	100.27	67.685	
N12_cation_cation_001	12	1	1	0.0	0.0	
N13_cation_cation_001	13	1	1	0.0	0.0	
N13_cation_cation_004	13	4	2	194.1	151.221	
N15_cation_cation_001	15	1	1	0.0	0.0	
N15_cation_cation_002	15	2	2	259.27	327.866	

Table 7: Classement DFT vs xTB, composés **anioniques**. Trié par rang DFT (ground state DFT = rang 1) au sein de chaque formule ; le rang xTB tel que généré à l'origine est rappelé pour voir si les deux classements concordent.

Nom	**N**	**rang xtb**	**rang DFT**	E_{rel} **xtb** (kcal/mol)	E_{rel} **DFT** (kcal/mol)	**accord GS**
N5_anion_anion_002	5	2	1	45.01	0.0	
N5_anion_anion_003	5	3	2	45.78	0.001	
N5_anion_anion_004	5	4	3	85.94	0.001	
N5_anion_anion_001	5	1	4	0.0	6.585	
N5_anion_anion_007	5	7	5	110.2	70.417	
N5_anion_anion_005	5	5	6	101.13	70.418	
N5_anion_anion_008	5	8	7	118.64	72.158	
N5_anion_anion_006	5	6	8	104.25	111.045	
N5_anion_anion_010	5	10	9	143.36	177.348	
N7_anion_anion_011	7	11	1	5.49	0.0	
N7_anion_anion_012	7	12	2	6.13	3.082	
N7_anion_anion_016	7	16	3	28.08	14.425	
N7_anion_anion_010	7	10	4	4.19	19.981	
N7_anion_anion_008	7	8	5	0.0	23.072	
N7_anion_anion_018	7	18	6	50.76	53.065	
N7_anion_anion_019	7	19	7	56.14	63.445	
N9_anion_anion_001	9	1	1	0.0	0.0	
N11_anion_anion_001	11	1	1	0.0	0.0	
N11_anion_anion_002	11	2	2	39.44	38.478	
N11_anion_anion_006	11	6	3	57.42	60.325	
N11_anion_anion_011	11	11	4	92.67	74.023	
N11_anion_anion_005	11	5	5	51.65	74.395	
N11_anion_anion_004	11	4	6	49.83	92.156	
N11_anion_anion_007	11	7	7	69.82	101.735	
N11_anion_anion_010	11	10	8	89.5	102.267	
N11_anion_anion_008	11	8	9	77.42	108.607	
N11_anion_anion_003	11	3	10	44.22	109.185	
N11_anion_anion_013	11	13	11	115.91	177.005	
N11_anion_anion_017	11	17	12	164.9	185.468	
N11_anion_anion_018	11	18	13	185.17	199.633	
N13_anion_anion_001	13	1	1	0.0	0.0	

4. Vérification des minima

Sur les 92 structures terminées avec un bloc de thermochimie complet, **19 présentent au moins une fréquence imaginaire** et ne sont donc pas, en l'état, de vrais minima locaux – concentrées de manière frappante sur N_{11}^{-} et N_{14} neutre. Contrairement au protocole gros-cluster, aucune redésymétrisation automatique n'est appliquée : chaque candidat flaggé devra être repris manuellement (distorsion le long du mode imaginaire puis réoptimisation) avant d'être retenu comme candidat final – exactement la logique de la colonne `is_true_minimum` / `n_freq_hops` déjà implémentée dans `polyN_pipeline.py` au niveau GFN2-xTB, appliquée ici au niveau DFT. Le détail (fréquence, intensité IR) est dans `results_imaginary_modes.csv`.

5. Orbitales frontières

Table 8: Orbitales frontières (WB97X-D4/aug-cc-pVTZ) des candidats terminés.

Nom	PG	HOMO (eV)	LUMO (eV)	Gap (eV)
N10_neutral_neutral_001	C2	-12.9198	-0.5398	12.3800
N10_neutral_neutral_002	C1	-11.7475	-1.3053	10.4422
N10_neutral_neutral_003	C1	-9.9826	-1.1691	8.8135
N10_neutral_neutral_004	C1	-9.7070	-1.4443	8.2627
N10_neutral_neutral_005	C1	-10.3189	-1.6633	8.6556
N10_neutral_neutral_006	Cs	-11.2573	-0.6247	10.6326
N10_neutral_neutral_007	Cs	-10.5916	-1.8512	8.7404
N10_neutral_neutral_010	C1	-11.1858	-0.6563	10.5295
N10_neutral_neutral_011	C1	-10.2546	-1.8095	8.4451
N10_neutral_neutral_013	C2v	-8.7777	-2.5200	6.2577
N10_neutral_neutral_014	C1	-10.1760	-3.1308	7.0452
N10_neutral_neutral_015	C2	-10.7038	-1.6499	9.0539
N10_neutral_neutral_016	C1	-10.7642	-1.5672	9.1970
N10_neutral_neutral_017	C2v	-11.2487	-1.3285	9.9202
N10_neutral_neutral_018	C2	-11.1371	0.6545	11.7916
N10_neutral_neutral_019	C3v	-11.3917	0.3446	11.7363
N10_neutral_neutral_020	C2	-10.5955	-1.4418	9.1537
N10_neutral_neutral_021	C2v	-11.2774	-1.0998	10.1776
N10_neutral_neutral_022	D5h	-11.4974	-0.8667	10.6307
N11_anion_anion_001	C1	-4.6868	4.7575	9.4443
N11_anion_anion_002	C1	-4.6917	4.4152	9.1069
N11_anion_anion_003	C2	-4.8000	3.0674	7.8674
N11_anion_anion_004	C1	-4.1785	4.4585	8.6370
N11_anion_anion_005	C1	-4.0097	4.1648	8.1745
N11_anion_anion_006	C1	-4.2089	4.0149	8.2238
N11_anion_anion_007	Cs	-4.6404	4.6470	9.2874
N11_anion_anion_008	C1	-3.9921	4.4067	8.3988
N11_anion_anion_010	Cs	-3.3157	4.1356	7.4513
N11_anion_anion_011	C1	-4.8669	2.8826	7.7495
N11_anion_anion_013	C1	-2.7734	2.6705	5.4439
N11_anion_anion_017	C1	-4.0058	4.4202	8.4260
N11_anion_anion_018	C1	-2.4542	3.0647	5.5189
N11_cation_cation_001	C2v	-16.4400	-6.4750	9.9650
N11_cation_cation_002	C1	-15.7017	-6.1480	9.5537
N11_cation_cation_005	C2	-17.0074	-7.0585	9.9489
N11_cation_cation_006	C1	-15.6724	-7.1994	8.4730
N12_neutral_neutral_001	C1	-12.3196	-2.4745	9.8451
N12_neutral_neutral_002	C1	-11.3338	-2.5898	8.7440
N12_neutral_neutral_003	C1	-11.9678	-1.5248	10.4430
N12_neutral_neutral_004	Cs	-9.8374	-2.7778	7.0596
N12_neutral_neutral_005	C1	-10.6336	-1.7561	8.8775

(suite)

Nom	PG	HOMO (eV)	LUMO (eV)	Gap (eV)
N12_neutral_neutral_006	C1	-10.3193	-1.8368	8.4825
N12_neutral_neutral_007	D3d	-11.6168	-0.4953	11.1215
N12_neutral_neutral_008	C1	-10.9868	-0.5129	10.4739
N12_neutral_neutral_009	D6h	-10.5865	-0.9025	9.6840
N13_anion_anion_001	C1	-4.2327	3.0401	7.2728
N13_cation_cation_001	C1	-15.9024	-7.4824	8.4200
N13_cation_cation_004	C1	-17.1447	-7.0618	10.0829
N14_neutral_neutral_001	C1	-9.4477	-2.1106	7.3371
N14_neutral_neutral_004	C1	-10.0443	-1.2482	8.7961
N14_neutral_neutral_005	C2	-12.0686	0.6111	12.6797
N15_cation_cation_001	C1	-16.9221	-7.4624	9.4597
N15_cation_cation_002	C1	-17.0792	-5.2422	11.8370
N4_neutral_neutral_001	Td	-13.9405	1.7935	15.7340
N4_neutral_neutral_002	C2v	-9.2289	-1.2933	7.9356
N4_neutral_neutral_003	C2v	-10.5771	-1.9734	8.6037
N4_neutral_neutral_004	D2h	-10.6441	-2.3507	8.2934
N5_anion_anion_001	D5h	-5.8838	5.8881	11.7719
N5_anion_anion_002	C1	-2.7204	5.1388	7.8592
N5_anion_anion_003	C1	-2.7207	5.1392	7.8599
N5_anion_anion_004	C1	-2.7202	5.1384	7.8586
N5_anion_anion_005	C1	-1.6987	5.2393	6.9380
N5_anion_anion_006	Cs	-1.8642	3.7995	5.6637
N5_anion_anion_007	C1	-1.6984	5.2388	6.9372
N5_anion_anion_008	C1	-1.5258	4.9909	6.5167
N5_anion_anion_010	C2v	-1.9078	5.4320	7.3398
N5_cation_cation_001	C2v	-18.4486	-6.5903	11.8583
N5_cation_cation_002	Cs	-19.4731	-7.2442	12.2289
N5_cation_cation_004	C1	-18.4488	-6.5881	11.8607
N5_cation_cation_005	D2d	-19.7750	-8.2030	11.5720
N5_cation_cation_006	C4v	-20.6577	-7.9402	12.7175
N6_neutral_neutral_001	C2	-9.2804	-0.0404	9.2400
N6_neutral_neutral_002	D2	-11.4121	-2.0522	9.3599
N6_neutral_neutral_003	C1	-10.1854	-0.1470	10.0384
N6_neutral_neutral_004	C1	-14.8605	1.5862	16.4467
N6_neutral_neutral_006	C1	-9.3271	-0.6904	8.6367
N6_neutral_neutral_007	C2	-12.1087	-0.8535	11.2552
N6_neutral_neutral_008	C2v	-12.0044	-0.7828	11.2216
N6_neutral_neutral_009	C2	-11.2453	-0.2179	11.0274
N6_neutral_neutral_010	C2v	-9.7889	-3.6564	6.1325
N6_neutral_neutral_012	D3h	-11.9779	0.1531	12.1310
N7_anion_anion_008	C1	-2.9901	4.4265	7.4166
N7_anion_anion_010	C2v	-4.2100	5.1843	9.3943
N7_anion_anion_011	C1	-2.2849	4.8177	7.1026
N7_anion_anion_012	C1	-2.2126	4.8717	7.0843
N7_anion_anion_016	C1	-3.5935	4.2032	7.7967
N7_anion_anion_018	Cs	-2.9551	4.2456	7.2007

(suite)

Nom	PG	HOMO (eV)	LUMO (eV)	Gap (eV)
N7_anion_anion_019	C1	-2.6908	5.1080	7.7988
N7_cation_cation_001	Cs	-16.8094	-7.0687	9.7407
N9_anion_anion_001	C1	-3.3287	4.4403	7.7690
N9_cation_cation_001	C1	-18.1169	-6.2607	11.8562

6. Charges atomiques

Charges de Mulliken et Löwdin, natives à ORCA. Les charges de Bader (QTAIM) et de Manz (DDEC6) restent reportées (§9) – ni l'une ni l'autre n'est native à ORCA.

Table 9: Charges atomiques (Mulliken et Löwdin) du meilleur minimum confirmé de chaque groupe (formule, charge) – min/max sur les atomes N. Charges de Bader/QTAIM non natives à ORCA, reportées (§9).

Nom	Mulliken min	Mulliken max	Löwdin min/max
N4_neutral_neutral_001	-0.000	0.000	-0.000 / 0.000
N5_anion_anion_002	-0.879	0.791	-0.792 / -0.019
N5_cation_cation_001	-0.409	0.929	-0.069 / 0.465
N6_neutral_neutral_001	-0.552	1.088	-0.280 / 0.279
N7_anion_anion_011	-0.844	1.113	-0.385 / 0.112
N7_cation_cation_001	-0.296	1.030	-0.060 / 0.447
N9_anion_anion_001	-0.818	1.443	-0.568 / 0.159
N9_cation_cation_001	-0.501	1.052	-0.105 / 0.448
N10_neutral_neutral_001	-0.282	0.818	-0.303 / 0.133
N11_anion_anion_006	-0.756	1.131	-0.375 / 0.197
N11_cation_cation_001	-0.656	1.068	-0.235 / 0.410
N12_cation_cation_001	-0.169	1.074	-0.269 / 0.207
N12_neutral_neutral_001	-0.319	1.408	-0.380 / 0.154
N13_anion_anion_001	-0.979	1.392	-0.394 / 0.224
N13_cation_cation_001	-0.293	0.835	-0.181 / 0.218
N14_neutral_neutral_005	-0.211	0.218	-0.121 / 0.073
N15_cation_cation_001	-0.422	0.873	-0.258 / 0.419

7. Distances et ordres de liaison N–N : corrélation avec ΔH_f

Pour les composés neutres, proportion de liaisons courtes (triple, double, ou délocalisées $\sim 1{,}5$) parmi les liaisons réelles, et ordre de Mayer moyen, en regard du ΔH_f GFN2-xTB. Tendance attendue : davantage de caractère multiple devrait corréler avec un ΔH_f plus élevé (structures plus contraintes, moins proches de 2 N_2) – à confirmer statistiquement une fois la campagne DFT complète (l'échantillon actuel, 45 structures neutres, montre la tendance qualitativement mais reste bruité).

Table 10: Composés **neutres** terminés : proportion de liaisons N–N courtes (triple/double/délocalisées) parmi les liaisons réelles (hors contacts non-liants), et ΔH_f GFN2-xTB. Seuils de classification en §9 de `harvest_results.py`.

Nom	**liaisons**	**BO Mayer moy.**	**% courtes**[1]	**classes présentes**	ΔH_f^{xtb}
`N10_neutral_neutral_001`	11	0.806	36%	delocalized 1.5, single	-134.2
`N10_neutral_neutral_002`	10	0.917	70%	delocalized 1.5, single, triple	-123.7
`N10_neutral_neutral_003`	9	1.071	67%	delocalized 1.5, single, triple	-110.7
`N10_neutral_neutral_004`	10	0.829	60%	delocalized 1.5, single, triple	-100.84
`N10_neutral_neutral_005`	10	0.937	50%	delocalized 1.5, double, single	-97.64
`N10_neutral_neutral_006`	12	0.792	25%	double, single	-95.64
`N10_neutral_neutral_007`	11	0.790	36%	delocalized 1.5, single	-80.55
`N10_neutral_neutral_010`	12	0.772	25%	delocalized 1.5, single	-45.58
`N10_neutral_neutral_011`	11	0.787	45%	delocalized 1.5, double, single	-41.25
`N10_neutral_neutral_013`	12	0.643	33%	delocalized 1.5, single	-34.65
`N10_neutral_neutral_014`	12	0.713	42%	delocalized 1.5, long/weak, single	-33.7
`N10_neutral_neutral_015`	12	0.768	25%	delocalized 1.5, single	-22.96
`N10_neutral_neutral_016`	12	0.760	25%	delocalized 1.5, double, single	-20.37
`N10_neutral_neutral_017`	13	0.733	15%	double, single	-3.74
`N10_neutral_neutral_018`	15	0.658	0%	single	8.82
`N10_neutral_neutral_019`	15	0.666	0%	single	13.77
`N10_neutral_neutral_020`	14	0.671	7%	delocalized 1.5, single	34.26
`N10_neutral_neutral_021`	13	0.781	15%	delocalized 1.5, single	40.4
`N10_neutral_neutral_022`	15	0.663	0%	single	49.74
`N12_neutral_neutral_001`	13	0.615	38%	delocalized 1.5, single	-147.92

(suite)

Nom	liaisons	BO Mayer moy.	% courtes	classes présentes	ΔH_f^{xtb}
N12_neutral_neutral_002	13	0.700	62%	delocalized 1.5, single	-134.68
N12_neutral_neutral_003	13	0.801	69%	delocalized 1.5, double, single	-112.01
N12_neutral_neutral_004	13	0.695	38%	delocalized 1.5, single	-102.34
N12_neutral_neutral_005	14	0.760	29%	double, single	-79.39
N12_neutral_neutral_006	14	0.722	29%	delocalized 1.5, double, single	-71.56
N12_neutral_neutral_007	18	0.645	0%	single	-46.96
N12_neutral_neutral_008	15	0.787	20%	delocalized 1.5, single	-14.76
N12_neutral_neutral_009	18	0.605	0%	single	94.51
N14_neutral_neutral_001	16	0.599	50%	delocalized 1.5, single	-140.21
N14_neutral_neutral_004	10	1.480	80%	delocalized 1.5, double, long/weak, single, triple	-38.65
N14_neutral_neutral_005	21	0.681	0%	long/weak, single	-6.87
N4_neutral_neutral_001	6	0.796	0%	single	27.32
N4_neutral_neutral_002	4	1.118	50%	delocalized 1.5, double, single	53.79
N4_neutral_neutral_003	5	0.841	0%	long/weak, single	58.33
N4_neutral_neutral_004	4	1.056	50%	delocalized 1.5, single	68.96
N6_neutral_neutral_001	5	1.248	80%	delocalized 1.5, single, triple	-63.71
N6_neutral_neutral_002	6	0.835	100%	delocalized 1.5	-39.62
N6_neutral_neutral_003	6	1.112	50%	delocalized 1.5, double, single, triple	-8.71
N6_neutral_neutral_004	3	2.336	100%	triple	-3.12
N6_neutral_neutral_006	6	1.133	67%	delocalized 1.5, long/weak, single, triple	15.09
N6_neutral_neutral_007	7	0.867	29%	delocalized 1.5, single	19.92
N6_neutral_neutral_008	8	0.790	12%	double, single	20.81
N6_neutral_neutral_009	7	0.979	29%	double, single	47.09

(suite)

Nom	liaisons	BO Mayer moy.	% courtes	classes présentes	ΔH_f^{xtb}
N6_neutral_neutral_010	6	0.935	33%	single, triple	50.62
N6_neutral_neutral_012	9	0.729	0%	single	72.95

8. Confrontation à la littérature et aux bases externes

Le fichier `biblio_polyN/comparison_table_3sources.csv` de `polyN-pipeline` catalogue 337 structures connues (articles publiés + bases externes `N_csp` et `polyN_study`, relaxées au même niveau GFN2-xTB). Le croisement par (N, charge) avec nos 193 candidats donne trois résultats complémentaires :

- **Couverture** : 12 des 19 groupes (89 des 193 structures, 46 %) n'ont **aucune** référence cataloguée à cette formule/charge.
- **Topologie** : pour les 104 candidats des 7 groupes comparables, une empreinte de graphe de liaisons (séquence de degrés + tailles de cycles + nombre de liaisons, seuil 1,90 Å, NetworkX) montre que 76 (73 %) ne correspondent à aucune connectivité cataloguée. Au total, **165 des 193 candidats (86 %)** sont donc distincts de tout ce qui existe dans le tableau à 3 sources.
- **Stabilité** : 4 des 7 groupes comparables voient leur meilleur candidat local dépasser en stabilité le meilleur catalogué – N_9^- ($-0{,}166$ eV/atome), N_9^+ ($-0{,}213$), N_{10} ($-0{,}066$), N_{12} ($-0{,}246$). N_6 et N_8 retombent exactement sur le minimum déjà connu. N_4 est la seule perte nette ($+0{,}302$ eV/atome).

Détail complet et graphique interactif : voir l'artefact *Uncharted Compositions*.

9. Limites et prochaines étapes

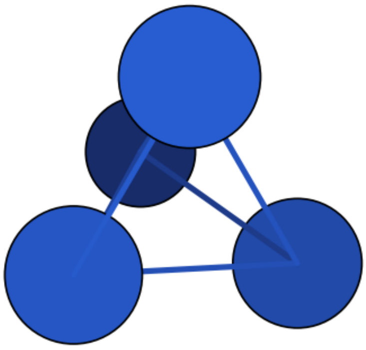
- La comparaison de stabilité de la section précédente reste au niveau **GFN2-xTB** – la confirmation définitive viendra des énergies WB97X-D4 elles-mêmes, une fois la campagne complète et les 19 candidats flaggés re-optimisés en vrais minima.
- Charges de Bader (QTAIM) et de Manz (DDEC6) reportées à une étape ultérieure – nécessitent respectivement le code de Henkelman et Chargemol sur un cube de densité via `orca_plot`, non installés sur le cluster à ce jour.
- Tous les fichiers ORCA bruts (`.gbw`, `.densities`, `.hess`) sont conservés intacts pour une seconde campagne CCSD(T) envisagée sur les petites structures – le module `refinement/` du dépôt compagnon `polyN` implémente déjà un enchaînement de méthodes config-driven avec seuil de taille (`max_n_atoms`) adapté à cet usage.
- Seuils de classification des liaisons N–N (§6) : triple $< 1{,}15$ Å, double $< 1{,}22$, délocalisée $< 1{,}32$, simple $< 1{,}55$, au-delà non-liant – purement géométriques, à recouper avec l'ordre de Mayer (fourni en regard dans `results_bonds.csv`) plutôt qu'à prendre isolément.

[1] Triple, double ou délocalisée (ordre $\sim 1{,}5$), à l'exclusion des liaisons simples et des contacts non-liants.

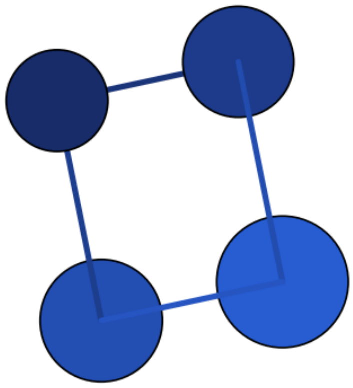
A. Structures optimisées (DFT, minima confirmés)

Les 73 candidats confirmés comme vrais minima au moment de la rédaction (aucune fréquence imaginaire), deux par ligne. Trois figures, classées par nombre d'atomes croissant puis par stabilité au sein de chaque formule : Figure S1 (neutres), Figure S2 (cations), Figure S3 (anions). Chaque structure porte son nom, son groupe ponctuel, son ΔH (kcal/mol, isomère le plus stable de sa formule $= 0$), le niveau de théorie de ce ΔH (DFT ou xTB si le DFT n'est pas encore terminé), sa référence (numéro d'article ou *our work*), et un indicateur si une distance N–N dépasse 1,7 Å.

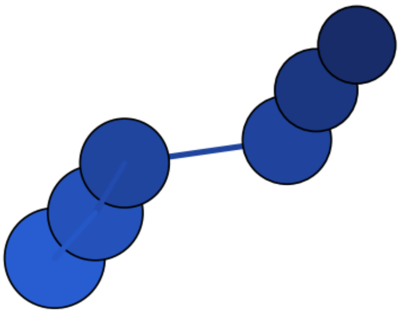
Figure S1 – composés neutres



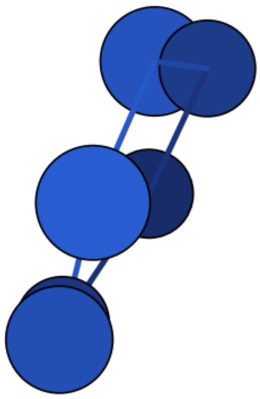
`N4_neutral_neutral_001` – Td – $\Delta H = 0.00$ (DFT) – our work



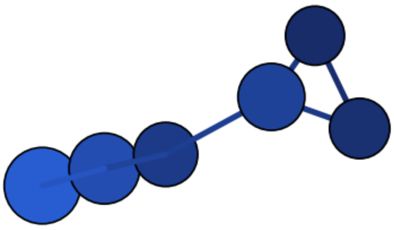
`N4_neutral_neutral_004` – D2h – $\Delta H = 3.38$ (DFT) – our work – N–N $> 1{,}7$ Å



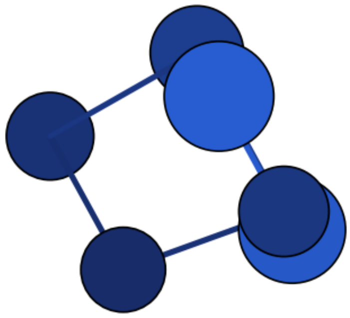
`N6_neutral_neutral_001` – C2 – $\Delta H = 172.76$ (DFT) – our work – N–N $> 1{,}7$ Å



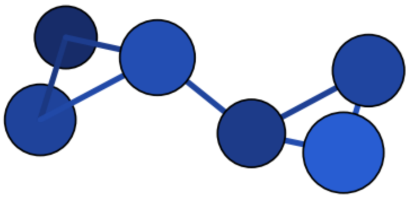
`N6_neutral_neutral_002` – D2 – $\Delta H = 195.83$ (DFT) – [1], our work – N–N $> 1{,}7$ Å



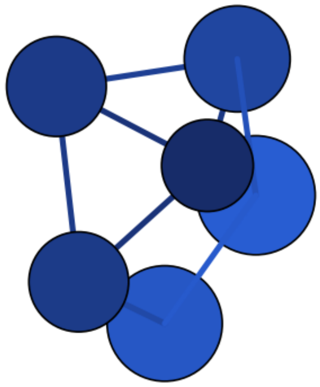
`N6_neutral_neutral_003` – C1 – $\Delta H = 200.83$ (DFT) – our work – N–N $> 1{,}7$ Å



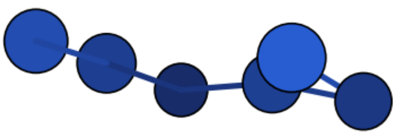
`N6_neutral_neutral_007` – C2 – $\Delta H = 226.12$ (DFT) – our work – N–N $> 1{,}7$ Å



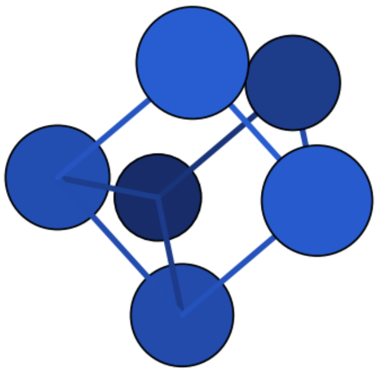
`N6_neutral_neutral_009` – C2 – $\Delta H = 231.01$ (DFT) – our work



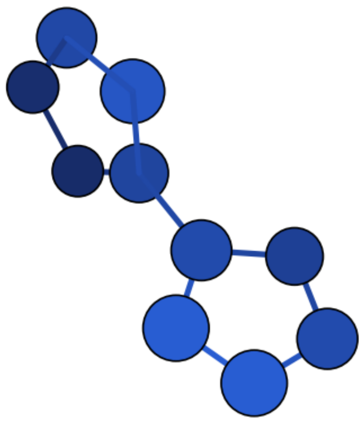
`N6_neutral_neutral_008` – C2v – $\Delta H = 236.66$ (DFT) – our work – N–N $> 1{,}7$ Å



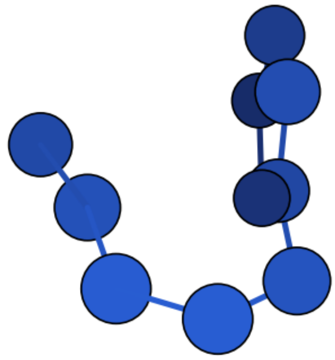
`N6_neutral_neutral_006` – C1 – $\Delta H = 258.25$ (DFT) – our work – N–N $> 1{,}7$ Å



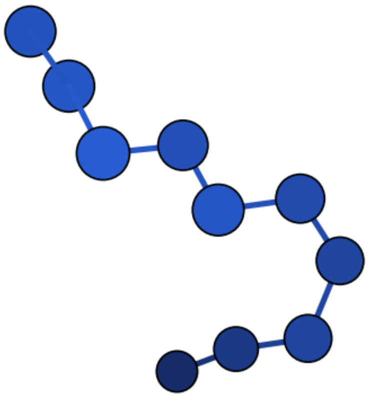
`N6_neutral_neutral_012` – D3h – $\Delta H = 305.23$ (DFT) – our work



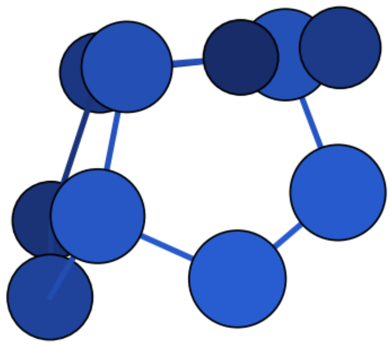
`N10_neutral_neutral_001` – C2 – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å



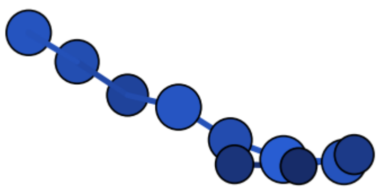
`N10_neutral_neutral_002` – C1 – $\Delta H = 30.61$ (DFT) – our work – N–N $> 1{,}7$ Å



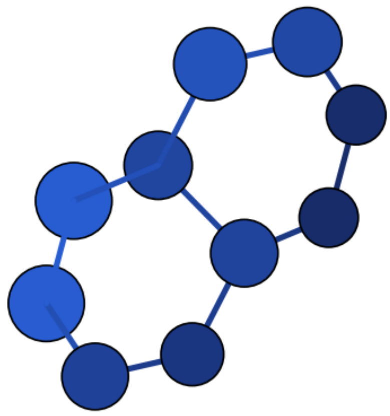
`N10_neutral_neutral_003` – C1 – $\Delta H = 50.60$ (DFT) – [1], our work – N–N $> 1{,}7$ Å



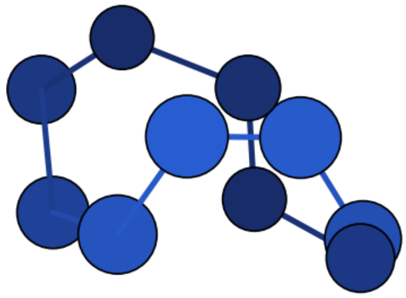
`N10_neutral_neutral_006` – Cs – $\Delta H = 69.56$ (DFT) – our work – N–N $> 1{,}7$ Å



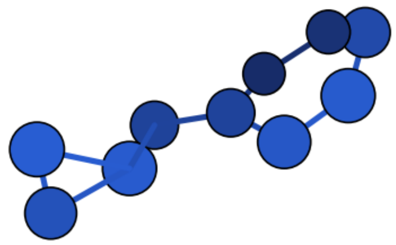
`N10_neutral_neutral_004` – C1 – $\Delta H = 71.13$ (DFT) – [1], our work – N–N $> 1{,}7$ Å



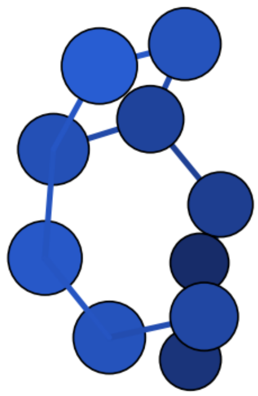
`N10_neutral_neutral_007` – Cs – $\Delta H = 87.34$ (DFT) – our work – N–N $> 1{,}7$ Å



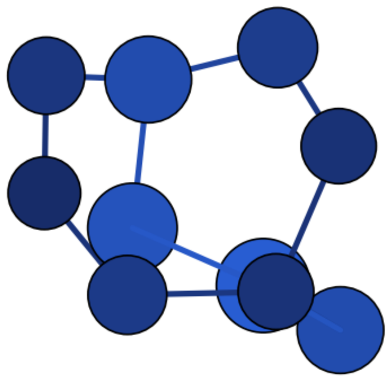
N10_neutral_neutral_005 – C1 –
$\Delta H = 102.77$ (DFT) – our work – N–N $> 1{,}7$ Å



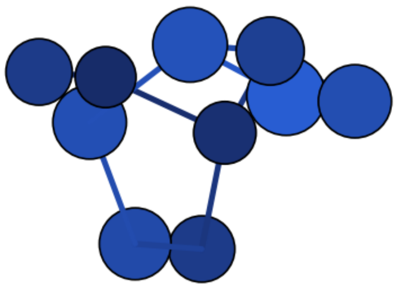
N10_neutral_neutral_011 – C1 –
$\Delta H = 107.23$ (DFT) – our work – N–N $> 1{,}7$ Å



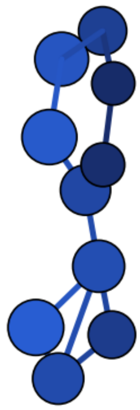
N10_neutral_neutral_010 – C1 –
$\Delta H = 118.99$ (DFT) – our work – N–N $> 1{,}7$ Å



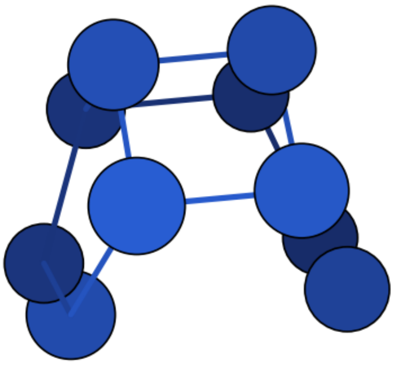
N10_neutral_neutral_015 – C2 –
$\Delta H = 123.40$ (DFT) – our work – N–N $> 1{,}7$ Å



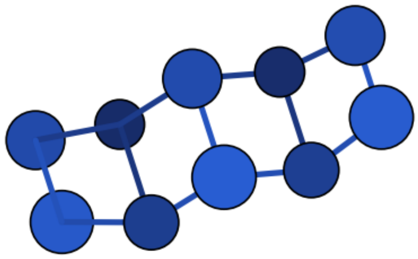
N10_neutral_neutral_016 – C1 –
$\Delta H = 126.63$ (DFT) – our work – N–N $> 1{,}7$ Å



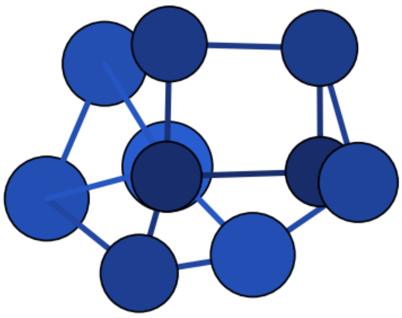
N10_neutral_neutral_014 – C1 –
$\Delta H = 139.29$ (DFT) – our work – N–N $> 1{,}7$ Å



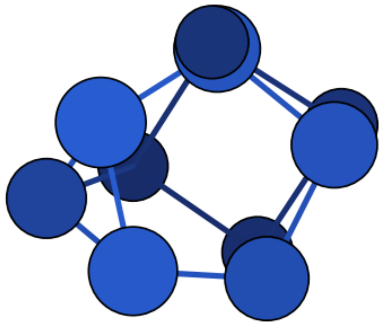
`N10_neutral_neutral_017` – C2v – $\Delta H = 151.14$ (DFT) – our work – N–N $> 1{,}7$ Å



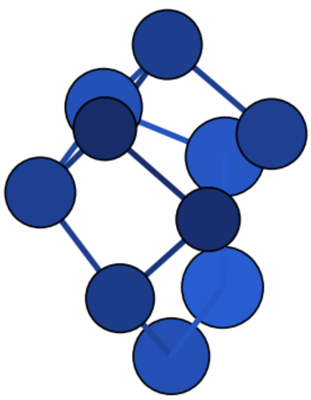
`N10_neutral_neutral_021` – C2v – $\Delta H = 173.84$ (DFT) – our work – N–N $> 1{,}7$ Å



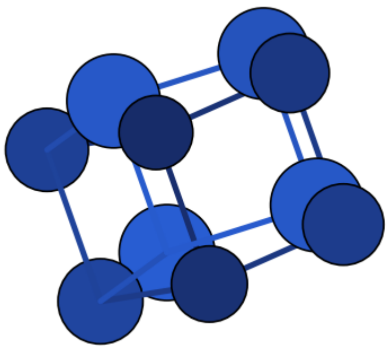
`N10_neutral_neutral_018` – C2 – $\Delta H = 199.07$ (DFT) – our work



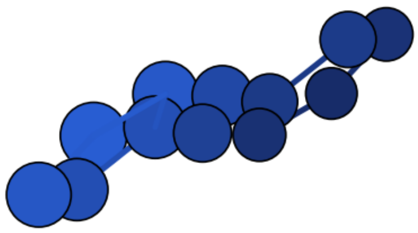
`N10_neutral_neutral_019` – C3v – $\Delta H = 203.80$ (DFT) – our work



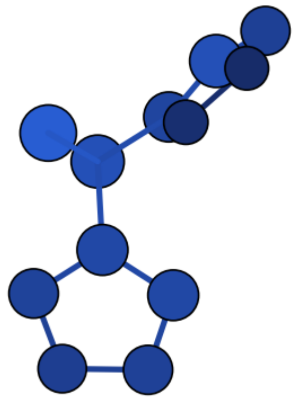
`N10_neutral_neutral_020` – C2 – $\Delta H = 214.21$ (DFT) – our work – N–N $> 1{,}7$ Å



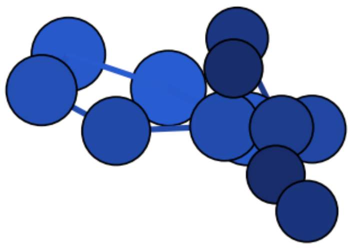
`N10_neutral_neutral_022` – D5h – $\Delta H = 241.14$ (DFT) – our work



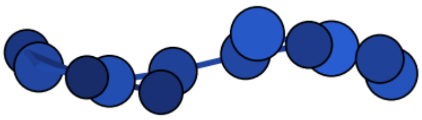
`N12_neutral_neutral_001` – C1 – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å



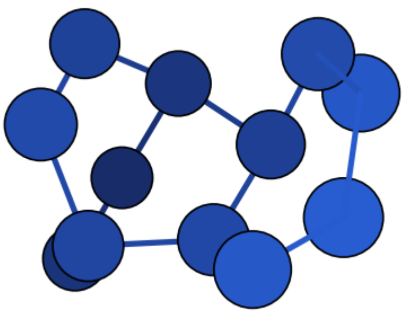
`N12_neutral_neutral_003` – C1 – $\Delta H = 36.41$ (DFT) – our work – N–N $> 1{,}7$ Å



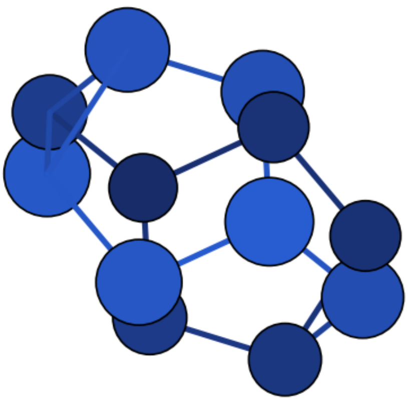
`N12_neutral_neutral_002` – C1 – $\Delta H = 56.99$ (DFT) – our work – N–N $> 1{,}7$ Å



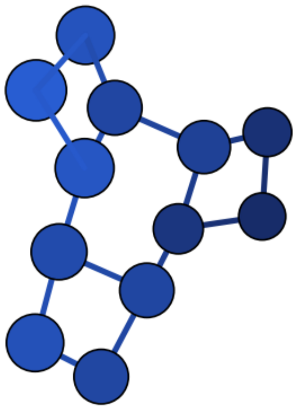
`N12_neutral_neutral_004` – Cs – $\Delta H = 116.12$ (DFT) – our work – N–N $> 1{,}7$ Å



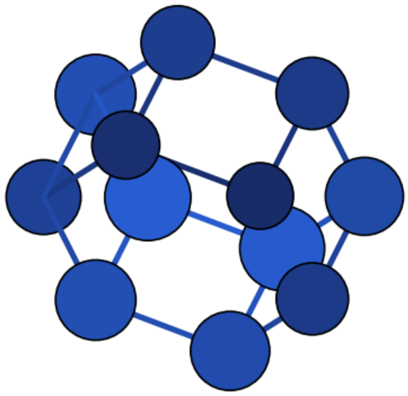
`N12_neutral_neutral_006` – C1 – $\Delta H = 127.50$ (DFT) – our work – N–N $> 1{,}7$ Å



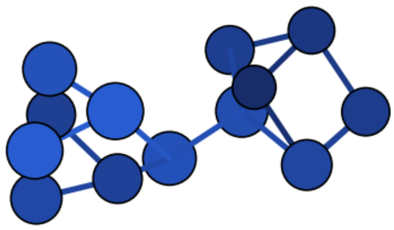
`N12_neutral_neutral_007` – D3d – $\Delta H = 188.33$ (DFT) – our work



`N12_neutral_neutral_008` – C1 – $\Delta H = 219.30$ (DFT) – our work – N–N $> 1{,}7$ Å

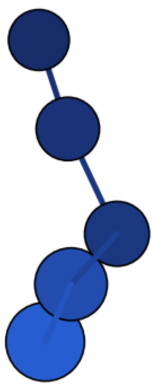


`N12_neutral_neutral_009` – D6h – $\Delta H = 334.92$ (DFT) – our work

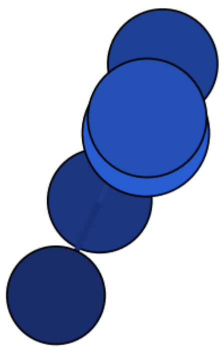


`N14_neutral_neutral_005` – C2 – $\Delta H = 341.35$ (DFT) – our work – N–N $> 1{,}7$ Å

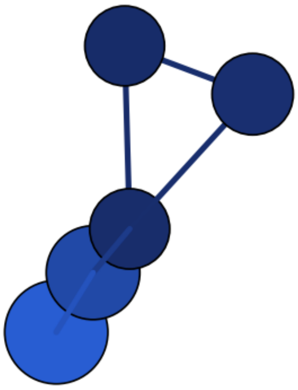
Figure S2 – composés cationiques



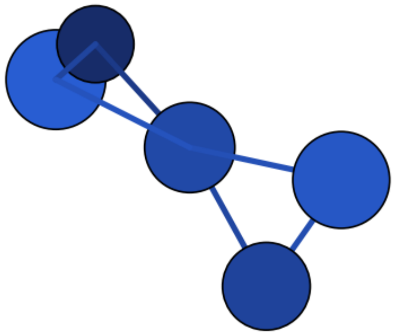
`N5_cation_cation_001` – C2v – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å



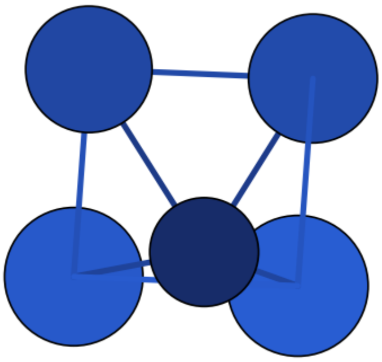
`N5_cation_cation_004` – C1 – $\Delta H = 0.01$ (DFT) – our work – N–N $> 1{,}7$ Å



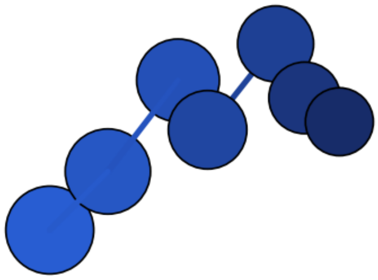
`N5_cation_cation_002` – Cs – $\Delta H = 41.63$ (DFT) – our work – N–N $> 1{,}7$ Å



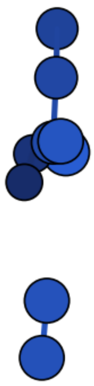
`N5_cation_cation_005` – D2d – $\Delta H = 109.53$ (DFT) – our work



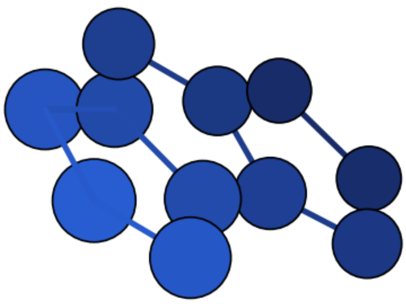
`N5_cation_cation_006` – C4v – $\Delta H = 155.74$ (DFT) – our work – N–N $> 1{,}7$ Å



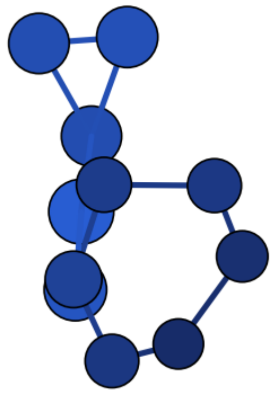
`N7_cation_cation_001` – Cs – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å



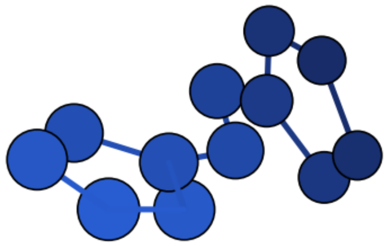
`N9_cation_cation_001` – C1 – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å



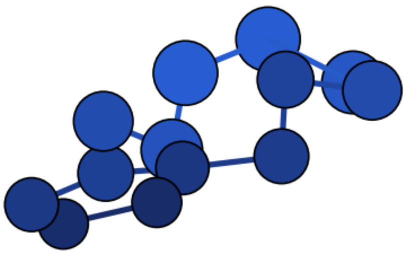
`N11_cation_cation_001` – C2v – $\Delta H = 23.30$ (DFT) – our work – N–N $> 1{,}7$ Å



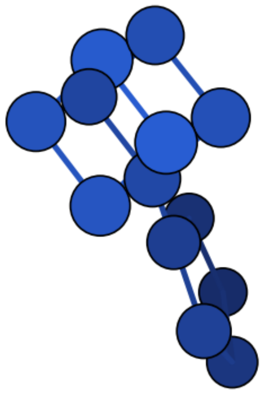
`N11_cation_cation_006` – C1 – $\Delta H = 67.68$ (DFT) – our work – N–N $> 1{,}7$ Å



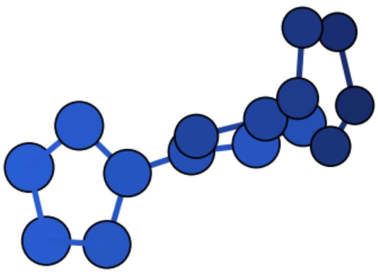
`N12_cation_cation_001` – C1 – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å



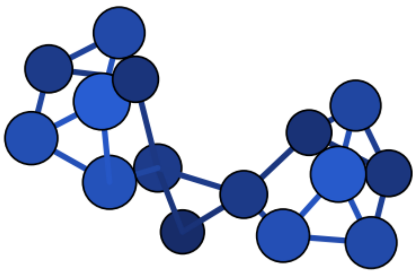
`N13_cation_cation_001` – C1 – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å



`N13_cation_cation_004` – C1 – $\Delta H = 151.22$ (DFT) – our work – N–N $> 1{,}7$ Å

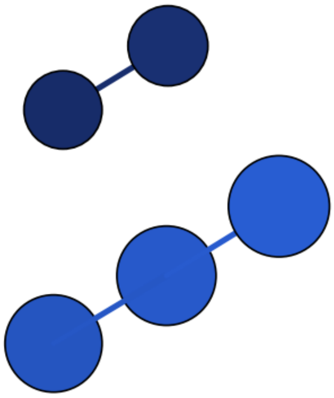


`N15_cation_cation_001` – C1 – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å

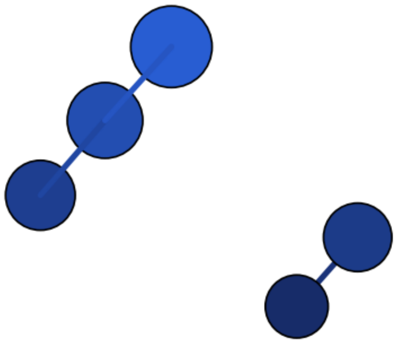


`N15_cation_cation_002` – C1 – $\Delta H = 327.87$ (DFT) – our work – N–N $> 1{,}7$ Å

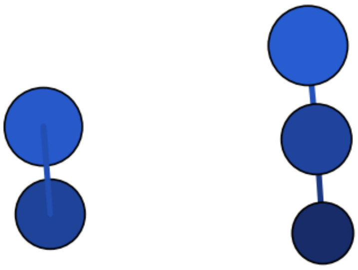
Figure S3 – composés anioniques



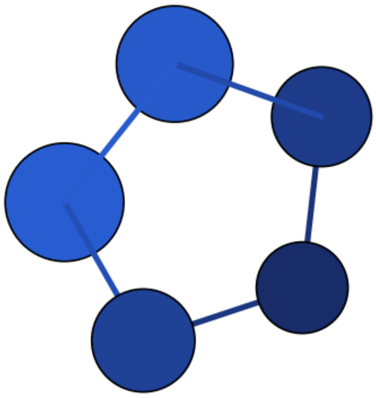
N5_anion_anion_002 – C1 – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å



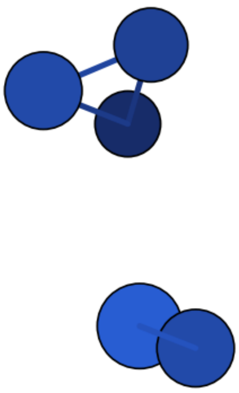
N5_anion_anion_003 – C1 – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å



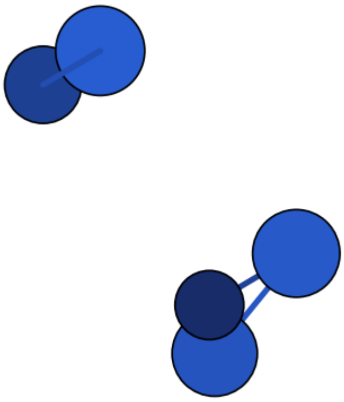
N5_anion_anion_004 – C1 – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å



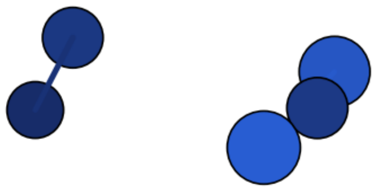
N5_anion_anion_001 – D5h – $\Delta H = 6.58$ (DFT) – our work – N–N $> 1{,}7$ Å



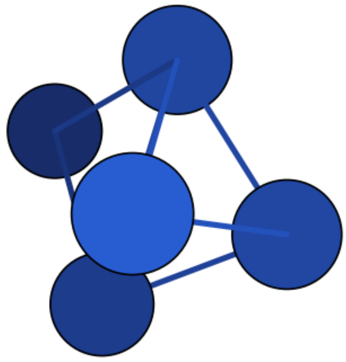
N5_anion_anion_007 – C1 – $\Delta H = 70.42$ (DFT) – our work



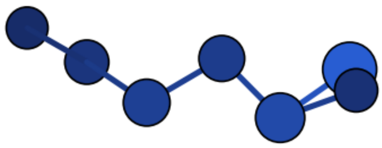
N5_anion_anion_005 – C1 – $\Delta H = 70.42$ (DFT) – our work



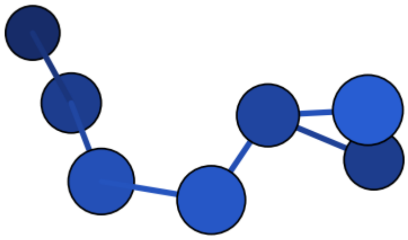
`N5_anion_anion_008` – C1 – $\Delta H = 72.16$ (DFT) – our work



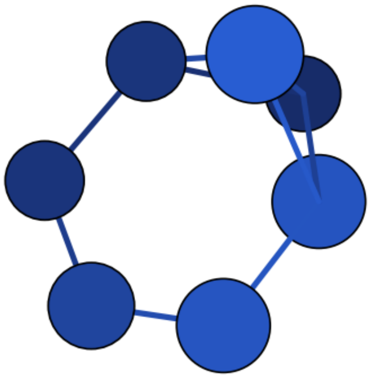
`N5_anion_anion_010` – C2v – $\Delta H = 177.35$ (DFT) – our work – N–N $> 1{,}7$ Å



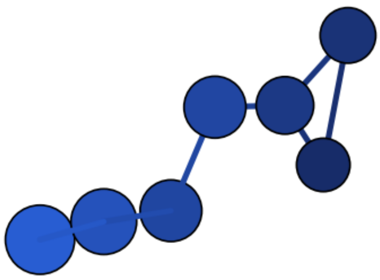
`N7_anion_anion_011` – C1 – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å



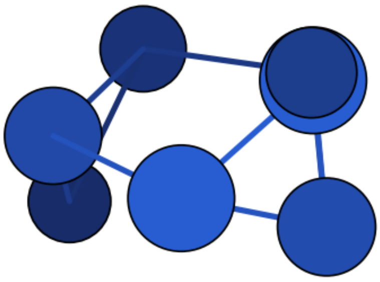
`N7_anion_anion_012` – C1 – $\Delta H = 3.08$ (DFT) – our work – N–N $> 1{,}7$ Å



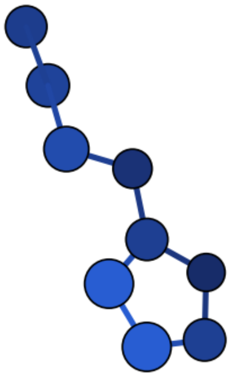
`N7_anion_anion_010` – C2v – $\Delta H = 19.98$ (DFT) – our work – N–N $> 1{,}7$ Å



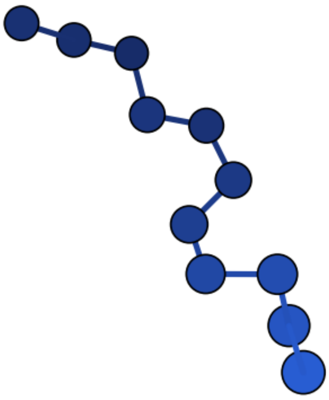
`N7_anion_anion_008` – C1 – $\Delta H = 23.07$ (DFT) – our work – N–N $> 1{,}7$ Å



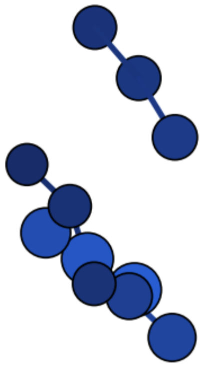
N7_anion_anion_019 – C1 – $\Delta H = 63.44$ (DFT) – our work



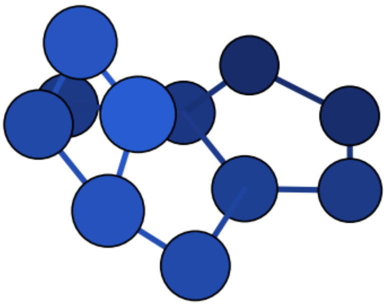
N9_anion_anion_001 – C1 – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å



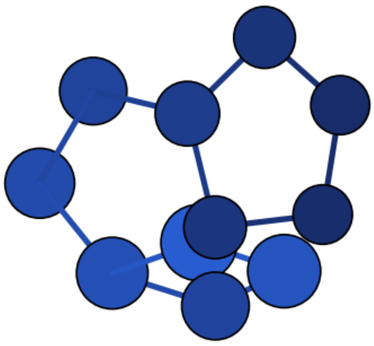
N11_anion_anion_006 – C1 – $\Delta H = 60.32$ (DFT) – our work – N–N $> 1{,}7$ Å



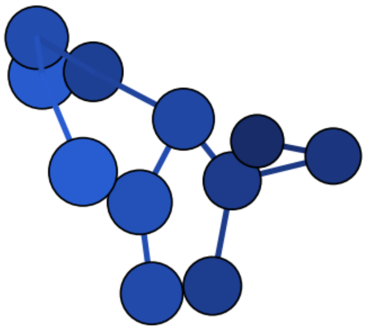
N11_anion_anion_011 – C1 – $\Delta H = 74.02$ (DFT) – our work – N–N $> 1{,}7$ Å



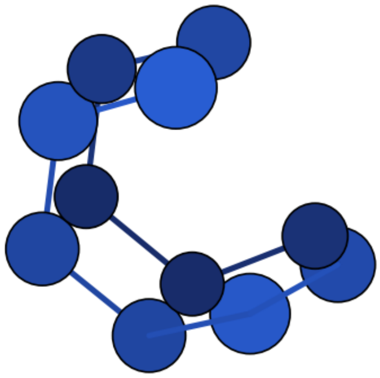
N11_anion_anion_007 – Cs – $\Delta H = 101.73$ (DFT) – our work – N–N $> 1{,}7$ Å



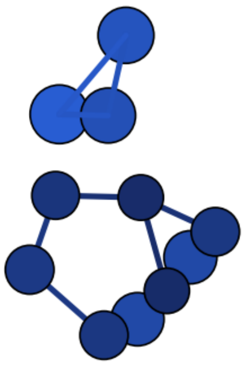
N11_anion_anion_008 – C1 – $\Delta H = 108.61$ (DFT) – our work – N–N $> 1{,}7$ Å



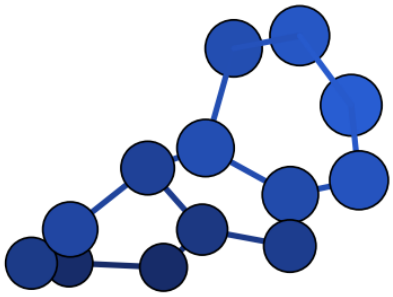
`N11_anion_anion_013` – C1 – $\Delta H = 177.01$ (DFT) – our work – N–N $> 1{,}7$ Å



`N11_anion_anion_017` – C1 – $\Delta H = 185.47$ (DFT) – our work – N–N $> 1{,}7$ Å



`N11_anion_anion_018` – C1 – $\Delta H = 199.63$ (DFT) – our work – N–N $> 1{,}7$ Å



`N13_anion_anion_001` – C1 – $\Delta H = 0.00$ (DFT) – our work – N–N $> 1{,}7$ Å

B. Références

Articles cités dans les tableaux de la section 2 (numérotation propre à ce rapport ; correspondance avec les codes de `biblio_polyN/table_references.tex` de `polyN-pipeline` en annotation).

[1] Fau, Mobita, Wilson, Perera, Bartlett, *Quantum Theory Project report*, University of Florida.

[2] Glukhovtsev, Jiao, von Ragé Schleyer, *Inorg. Chem.* **1996**, 35, 7124–7133.